



JURONG JUNIOR COLLEGE
2017 JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CLASS

17S

EXAM INDEX

CHEMISTRY

9729/03

Paper 3 Free Response

12 September 2017

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

A *Data Booklet* is provided.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This document consists of **11** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1 (a) (i) Describe the structure and bonding in Group 2 elements. Hence, explain why the melting point of calcium is higher than that of barium. [2]
- (ii) Samples of calcium and barium are separately added to beakers of cold water containing a few drops of litmus solution.
- Describe one observation you would make
- with calcium and barium
 - with calcium only [2]
- (b) Explain why the first ionisation energy of calcium is higher than that of potassium and gallium. [2]
- (c) Using inert electrodes, electrolysis of two separate solutions containing dilute and concentrated barium chloride respectively gives different products at the anode.
- Quote suitable E° values in the *Data Booklet* and use these values to deduce the product formed at the anode for each case. Explain your answer. [4]
- (d) The salts of the Group 2 elements with divalent anions (i.e. those with a charge of 2–) usually show a decreasing trend in solubility down the group. On the other hand, most of their salts with monovalent anions are very soluble.
- Magnesium hydroxide, $\text{Mg}(\text{OH})_2$, is almost insoluble in water. Stirring a mixture of magnesium hydroxide and aqueous ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$, produces a clear colourless solution containing **A**. When a solution of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, is added, a white precipitate **B** is formed.
- (i) Identify **A** and **B** and hence write balanced equations for the reactions. [2]
- The solubility product, K_{sp} , of $\text{Mg}(\text{OH})_2$ at 298 K has a numerical value of 2.00×10^{-11} .
- (ii) Write an expression for the solubility product, K_{sp} , of $\text{Mg}(\text{OH})_2$, stating its units. [2]
- (iii) Some $\text{Mg}(\text{OH})_2$ solid is added to pure water to obtain a saturated solution. Use the value of K_{sp} to calculate the concentration of $\text{Mg}(\text{OH})_2$ in a saturated solution and hence its pH. [2]

- (e) The enthalpy change of solution for Mg(OH)_2 , $\Delta H_{\text{sol}}(\text{Mg(OH)}_2(\text{s}))$ is represented by the following equation.



Using relevant data from the *Data Booklet* and the given data in the table below, construct a fully labelled energy cycle to calculate a value for the enthalpy change of solution of Mg(OH)_2 , $\Delta H_{\text{sol}}(\text{Mg(OH)}_2(\text{s}))$.

enthalpy change	value / kJ mol^{-1}
$\Delta H_{\text{f}}(\text{Mg(OH)}_2(\text{aq}))$	−922
$\Delta H_{\text{f}}(\text{OH}^{-}(\text{g}))$	−133
lattice energy of Mg(OH)_2	−2993
enthalpy change of atomisation of $\text{Mg}(\text{s})$	+148

[4]

[Total: 20]

- 2 (a) Chlorine, bromine and iodine all react with hydrogen.

With reference to suitable bond energies data in the *Data Booklet*, state and explain the trend in the reactivities of the halogens with hydrogen.

[2]

- (b) (i) When aqueous bromine is added to aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, a sulfur-containing product is formed. Upon addition of aqueous barium nitrate, a white precipitate appears which is insoluble in excess dilute nitric acid.

Identify the white precipitate.

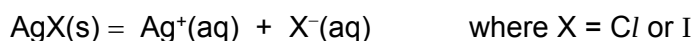
[1]

- (ii) However, no white precipitate is formed when the same experiment in (b)(i) is repeated with aqueous iodine instead of bromine. The sulfur-containing product formed is sodium tetrathionate, $\text{Na}_2\text{S}_4\text{O}_6$.

By considering the change in oxidation state of sulfur, explain the difference between the reactions of the two halogens with $\text{S}_2\text{O}_3^{2-}$.

[2]

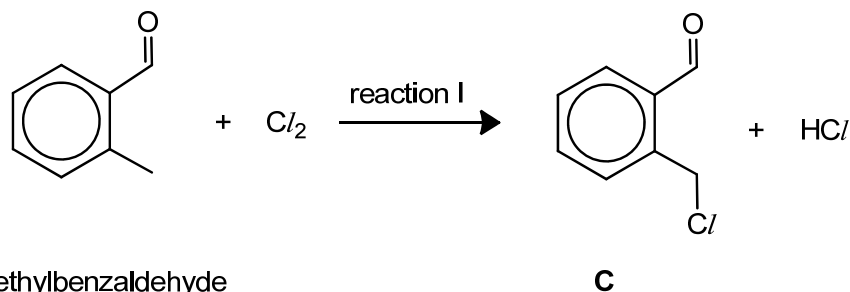
- (c) Silver chloride, AgCl , and silver iodide, AgI , are sparingly soluble in water.



Upon addition of excess aqueous ammonia, AgCl dissolves completely but AgI hardly dissolves. Explain these observations.

[2]

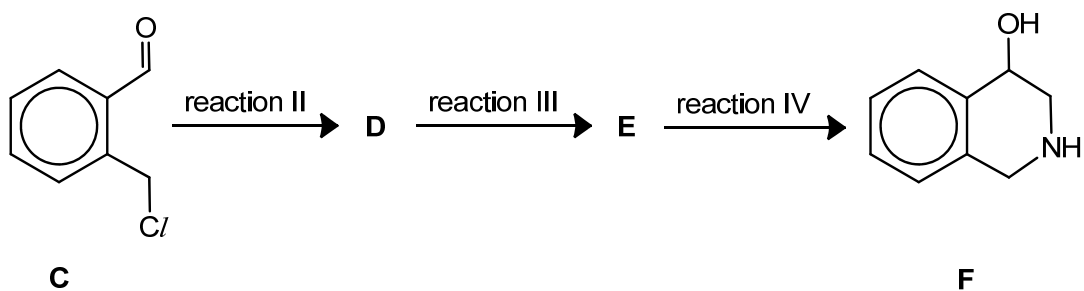
- (d) When reacted with chlorine under suitable conditions, 2-methylbenzaldehyde forms compound **C**.



- (i) Name and describe the mechanism for reaction I. You may show all the bonds in the methyl group in your mechanism.

[4]

C undergoes further reactions to form compound **F**.



(ii) Suggest reagents and conditions for reactions II to IV. [3]

(iii) Draw the structures of **D** and **E**. [2]

(iv) Alkane **G**, C_6H_{14} , undergoes a similar reaction as 2-methylbenzaldehyde with chlorine and produces **only three** monochlorinated constitutional isomers, **H**, **J** and **K**.

H exhibits stereoisomerism whereas **J** and **K** do not.

Suggest the type of stereoisomerism present in **H** and explain what feature is present in **H** that gives rise to this stereoisomerism.

Suggest the structures of **G** and **H**. [4]

[Total: 20]

- 3 The name of the element, *chromium*, means colour in Greek because many chromium compounds are intensely coloured.

(a) State the full electronic configuration of a chromium atom in its ground state. [1]

(b) Chromium ions tend to form octahedral complexes.

In an octahedral complex, ligands cause the five degenerate *d* orbitals of the chromium ion to be split into two energy levels.

(i) Using Cartesian axes, like those shown in Figure 3.1, draw **fully-labelled** diagrams of the following.

- One of the *d* orbitals at the lower energy level in an octahedral complex. Label this diagram as “**lower energy**”.
- One of the *d* orbitals at the higher energy level in an octahedral complex. Label this diagram as “**higher energy**”.

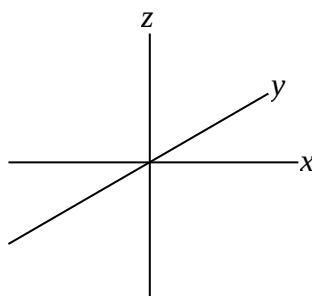


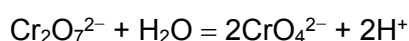
Figure 3.1

[2]

(ii) Hence, explain how the colour of chromium complexes arises. [2]

(c) Due to its ability to vary its oxidation state, chromium compounds tend to undergo redox reaction readily.

$\text{Cr}_2\text{O}_7^{2-}$ and CrO_4^{2-} ions are the oxoanions of chromium in its highest oxidation state of +6. In aqueous solution, $\text{Cr}_2\text{O}_7^{2-}$ and CrO_4^{2-} ions are interconvertible since they exist in a chemical equilibrium as shown below.



(i) In acidic medium, the predominant species is $\text{Cr}_2\text{O}_7^{2-}$ while in alkaline medium, it is CrO_4^{2-} .

Explain why, in alkaline medium, the predominant species is CrO_4^{2-} . [1]

(ii) Both $\text{Cr}_2\text{O}_7^{2-}$ and CrO_4^{2-} ions are moderately strong oxidising agents. When reduced, green Cr^{3+} solution was formed in acidic medium but in alkaline medium, grey-green $\text{Cr}(\text{OH})_3$ precipitate was formed.

Write an ion–electron equation for the reduction of CrO_4^{2-} to $\text{Cr}(\text{OH})_3$. [2]

- (iii) When a solution of CrO_4^{2-} is reacted with 0.300 mol of an appropriate reducing agent, 0.200 mol of $\text{Cr}(\text{OH})_3$ was formed.

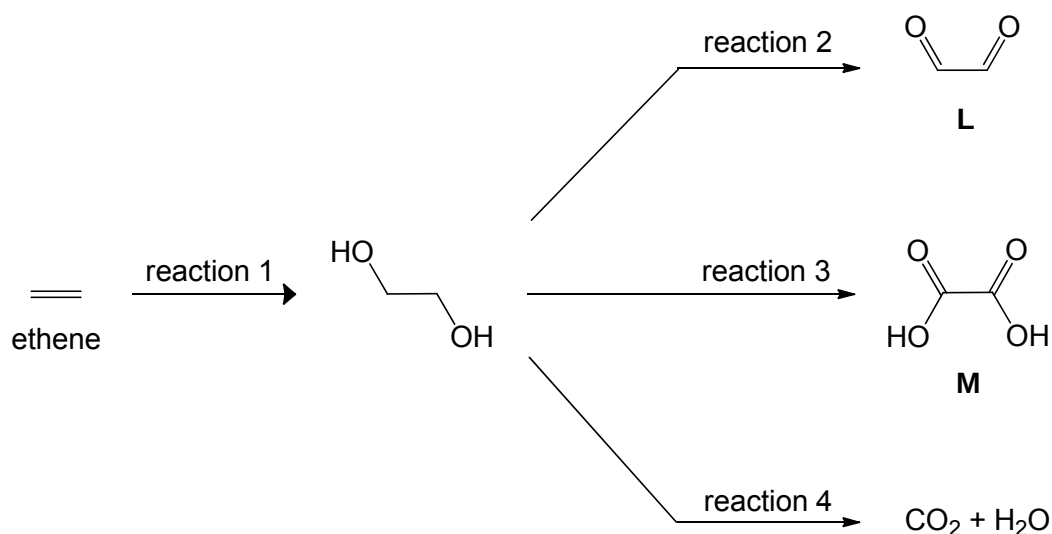
Using your answer in (c)(ii), together with the following data, calculate E_{cell} and standard electrode potential for $\text{CrO}_4^{2-}/\text{Cr}(\text{OH})_3$.

standard Gibbs free energy change of the reaction, $\Delta G^\circ = -365 \text{ kJ mol}^{-1}$

standard electrode potential of the reducing agent $= -0.76 \text{ V}$ [3]

- (iv) Hence, using relevant data from the *Data Booklet*, explain how this value in (c)(iii) relates to the relative oxidising power of the two chromium(VI) compounds. [1]

- (d) Similar to $\text{Cr}_2\text{O}_7^{2-}$, compounds containing MnO_4^- ions are commonly used in organic synthesis as good oxidising agents.

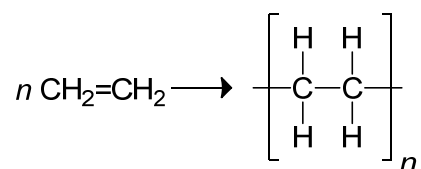


- (i) State the necessary reagents and condition for reaction 1 to 4. [4]

- (ii) A sodium salt of **M** was added to an aqueous solution of CrCl_3 , and a six-coordinate complex ion, $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, was formed.

Draw the structure of this complex ion, showing clearly the shape around chromium. [1]

- (e) In the production of approximately half of the world's polyethylene, a chromium oxide containing catalyst, also known as the Phillips catalyst, was used in the polymerisation of ethene.



State the **type** of catalysis involved in this reaction.

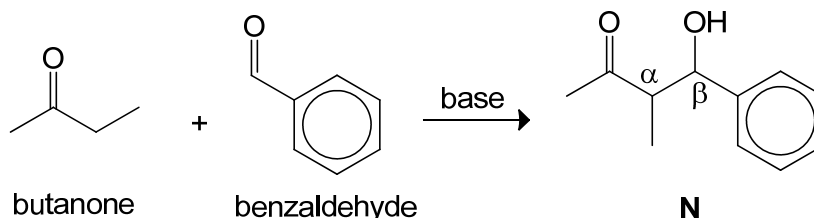
Describe how this **type** of catalyst speeds up the reaction.

[3]

[Total: 20]

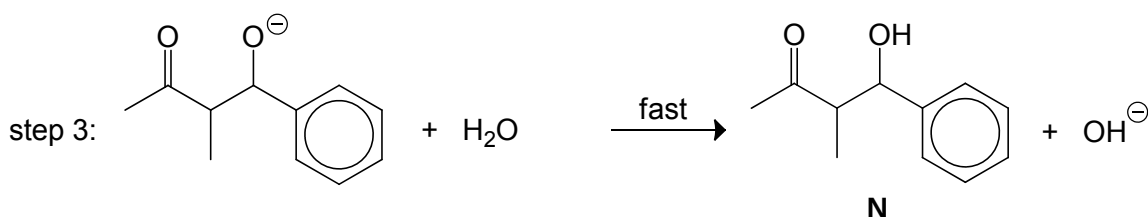
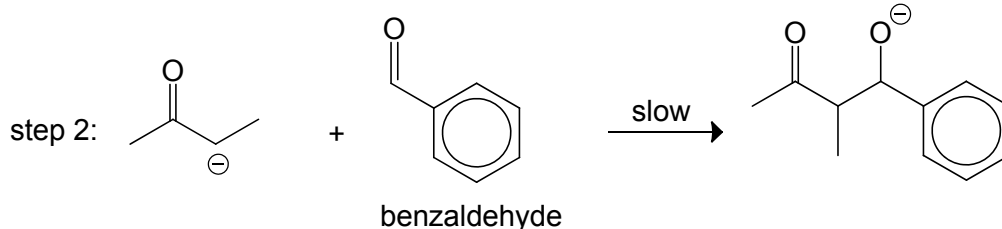
4 The Aldol Addition reaction is important in organic synthesis as it provides a method for linking two smaller molecules by introducing a carbon-carbon bond between them.

One example of Aldol Addition reaction is shown in the equation below.

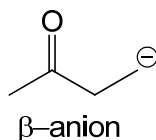


step 1:  + OH^- $\xrightarrow{\text{fast}}$  + H_2O

butanone α -anion

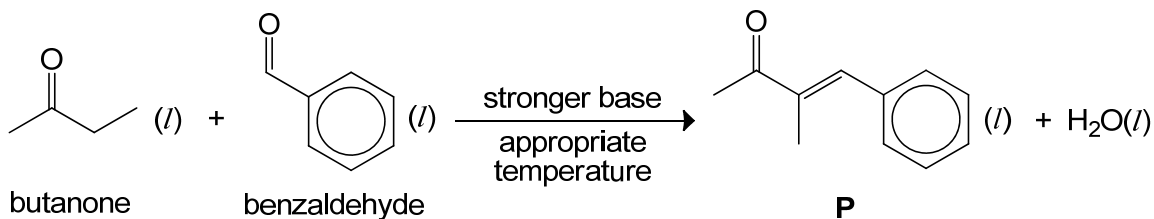


- (a) (i) State the type of reaction that has occurred during step 1 of the mechanism. [1]
- (ii) By considering the stability of the anion, explain why the α -anion is predominantly formed in step 1, instead of the β -anion as shown below.

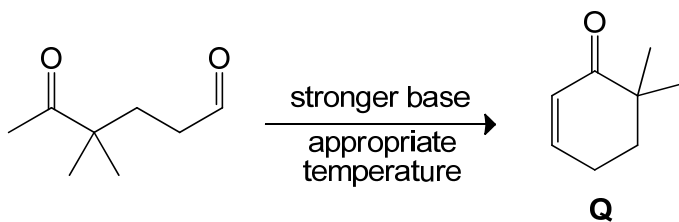


- (iii) State the role of the α -anion in step 2 of the mechanism. [1]
- (iv) Copy the equations for steps 2 and 3 shown above and complete the mechanism by showing the movement of electron pairs using curly arrows. Indicate all charges and relevant lone pairs. [2]

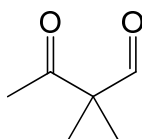
By either adjusting the reaction temperature or using a stronger base, butanone and benzaldehyde can undergo Aldol Condensation reaction, forming compound **P** as shown below.



- (b) (i) Using bond energies data in the *Data Booklet*, calculate the enthalpy change for this Aldol Condensation reaction. [2]
- (ii) The actual enthalpy change for the Aldol Condensation reaction between butanone and benzaldehyde is **much** more endothermic than that in (b)(i). Suggest **two** reasons to account for this difference in value. [2]
- (iii) Given that there is an increase in the entropy of the reaction, state and explain whether the Aldol Condensation reaction is favoured by a low or high reaction temperature. [2]
- (iv) Suggest a simple chemical test that could be carried out to see if compound **P** is formed. [2]
- (c) (i) State the Lewis theory of acids and bases. [1]
- (ii) Predict, with reasoning, which base, OH^- or $\text{CH}_3\text{CH}_2\text{O}^-$, would favour the Aldol Condensation reaction. [2]
- (d) Intramolecular Aldol Condensation reaction can also occur to give a cyclic unsaturated carbonyl compound when a dicarbonyl compound with a sufficiently long carbon chain is used. An example of such a reaction is given below.



- (i) State a value for the C=C–C bond angle in **Q**. [1]
- (ii) Hence suggest why the expected cyclisation does **not** occur when the following dicarbonyl compound was used.

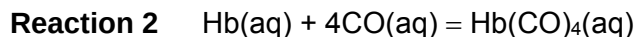
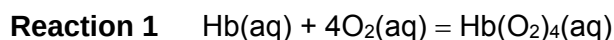


- (iii) Hence draw the structure of the unsaturated carbonyl product formed when the compound in (d)(ii) was used. [1]

[Total: 20]

- 5 An oxocarbon is a chemical compound consisting of only carbon and oxygen atoms, with carbon monoxide and carbon dioxide being the most common oxocarbons.

- (a) Carbon monoxide is one of the most widespread and dangerous industrial hazards due to the formation of carboxyhaemoglobin, $\text{Hb}(\text{CO})_4$, which occurs much more readily than the formation of oxyhaemoglobin, $\text{Hb}(\text{O}_2)_4$.



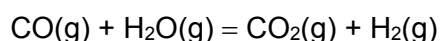
- (i) Draw separate dot-and-cross diagrams for carbon monoxide and oxygen.

Hence, explain why the formation of carboxyhaemoglobin occurs much more readily than that of oxyhaemoglobin [3]

- (ii) Treatment of severe carbon monoxide poisoning requires oxygen therapy which provides pure oxygen to the patient.

With reference to the equations given above, explain how oxygen therapy works in the treatment of carbon monoxide poisoning [2]

- (b) The water–gas shift reaction (WGS) is an industrial method to produce hydrogen gas, by reacting carbon monoxide with water vapour, which is an important reagent in industrial processes such as the production of ammonia gas in Haber process.



- (i) State two assumptions of the kinetic theory of an ideal gas. [2]

- (ii) A 500 cm^3 evacuated vessel was initially filled with $2.0 \times 10^{-4} \text{ mol}$ of $\text{CO}(\text{g})$ and $4.0 \times 10^{-4} \text{ mol}$ of $\text{H}_2\text{O}(\text{g})$ at 600 K . The vessel was sealed and left to stand. At equilibrium, the percentage yield of H_2 was found to be 97% .

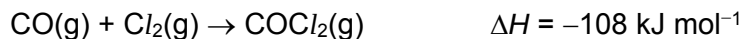
Using the ideal gas equation, calculate the equilibrium partial pressure of H_2 . [1]

- (iii) Comment whether the use of the ideal gas equation in (b)(ii) is valid. [1]

- (iv) Write an expression for the equilibrium constant, K_p , for the WGS reaction and state the units. [2]

- (v) Using your answer in (b)(ii), calculate a value of K_p for the WGS reaction that occurred at 600 K , expressing your answer to **two** significant figures. [2]

- (c) Phosgene, COCl_2 , is a valued industrial reagent used in the synthesis of pharmaceuticals and other organic compounds. It can be produced by passing purified carbon monoxide and chlorine gas through a bed of porous activated carbon, which serves as a catalyst.



Typically, the reaction is conducted between 50 °C and 150 °C. Above 200 °C, phosgene reverts to carbon monoxide and chlorine.

- (i) Predict and explain the sign of ΔS° for the production of phosgene. [1]
- (ii) Use your answer in (c)(i) and the information given, deduce whether the production of phosgene is enthalpy driven or entropy driven. [1]
- (iii) Phosgene is sensitive towards moisture and will give an acidic solution in the presence of moisture. Write an equation to illustrate this reaction. [1]
- (d) When reacted with aqueous sodium carbonate, 1 mole of an aromatic compound **R**, $\text{C}_{12}\text{H}_6\text{O}_{12}$, produces 72 dm³ of carbon dioxide measured at room temperature and pressure.
- Heating **R** with appropriate reagent for several hours produces a sweet-smelling compound **S**, C_{12}O_9 , which is symmetrical.
- Deduce the structures for compounds **R** and **S**, explaining the chemistry of the reactions involved. [4]

[Total: 20]

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